



Examiners' Report

Summer 2026

Pearson Edexcel GCE

in Further Mathematics (9FM0)

Paper 01 Core Pure Mathematics 1

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General Comments

The paper was accessible to learners across the ability range, with good entry points and opportunities for learners to demonstrate secure understanding of key methods. Routine tasks were generally approached successfully, while questions requiring greater rigour, careful algebraic manipulation or interpretation provided effective discrimination.

Presentation of working was generally sufficient for examiners to follow learners' methods. However, some responses, particularly in Question 7, would have benefited from a clearer structure and more systematic algebraic communication.

Comments on individual questions

Question 1

Question 1 provided an accessible opening to the paper. Most learners were able to demonstrate understanding of the matrix transformation, and many responses showed a sound grasp of how to interpret the transformation geometrically.

Part (a) was very well answered by most learners, with most understanding the need the transformation matrix by the column method and doing so correctly. Despite the question asking for the coordinates of the point, many responses left their answer as a column vector, which was an acceptable answer even though not in a coordinate form. Only a small number of learners attempted to multiply in the incorrect order, and generally this attempt was not awarded marks. Algebraic slips on the y coordinate, or other minor errors, were also occasionally seen.

Again, part (b) was also done well by most learners, with most able to set up the required equation and rearrange to show the required result. Most recognised all they need to do was to substitute their coordinates into $x + y = 4$ and simplify, but there were also many who used a matrix method in (b) rather than direct substitution into the given equation. Again, these attempts were mostly successful, though were slightly more prone to error.

The majority who attempted part (c) in the question were able to gain the mark. Most recognised the transformation was a rotation of 60° , and the mark was

awarded in all cases where 60° (or equivalent angle in radians) was stated even if the full description was not given. Others recognised it was a rotation about an axis but were not successful in finding an angle. Indeed, the mark was not accessible for those who did not attempt the question with any relevant work or who did not give the correct angle. For example, there were some learners who stated an incorrect transformation (such as rotation in a plane or stretch/enlargement about the origin) and did not give an angle, while a few solved $2 \sec \theta = 4$ to give 30° or some other angle instead of the 60° .

Question 2

This question was generally well answered. Learners demonstrated a good understanding of complex conjugates, manipulation of real and imaginary parts of complex numbers and their representation on an *Argand* diagram.

In part (a), errors were rare. Most learners correctly identified the complex conjugate and proceeded to show the given result. As this was a 'show that' question, accuracy in algebraic manipulation and the clear presentation of intermediate steps were essential. The most common omission was not attempting to explicitly simplify $4i^2$ to -4 before reaching the printed result, sometimes resulting in a sign error with some answers showing $+4$ instead of -4 . There were also occasional algebraic errors when working with powers of λ .

Part (b) was also answered well. A common error was setting the imaginary part equal to zero rather than the real part. Most learners who obtained both values of λ correctly rejected $\lambda = -4$, although justification was not always clearly shown.

Most answers correctly determined $|z|$, with the follow through mark allowing even those who started with a different equation in part (b) scoring this mark by following through from their value of λ . In part (ii) most could identifying the relevant arctangent value needed for the principal argument, but there were some errors when selecting an angle in the wrong quadrant, usually by not considering the negative imaginary part or by rotating the principal argument in different direction. Other errors included evaluating $\arctan\left(\pm \frac{1}{2}\right)$ instead of using the correct ratio or giving an answer of -1.12 rather than -1.11 due to rounding it earlier or missing the negative sign and giving $+1.11$ instead of -1.11 .

The Argand diagram for part (d) was usually well drawn. Most learners correctly plotted z , z^* , and $3z^* + z^2$ with relative scales and labelling showing the

conjugate pair symmetrically about the real axis and recognising that $3z^* + z^2$ lay on the imaginary axis at the same height as z^* . Common errors included plotting $3z^* + z^2$ on the real axis instead of the imaginary axis, reversing the positions of z and z^* or plotting $3z^*$ and z^2 as separate points

Question 3

This was a routine question in which well-structured responses were often successful, particularly in part (a).

Part (a) score well, with the standard formulae applied correctly and the expression expanded and factorised correctly to achieve the result. Expansion of the brackets was done well with only very occasional instances of incorrect signs or coefficients. The standard summation formulae were used correctly in most cases with only occasional instances where $\sum_{r=1}^n 1 = n$ was not used (an error that meant that the terms did not have a common factor of n and so the rest of the marks were not awarded for this part).

Most answers proceeded then to expand the expression fully before cancelling and factorising to the required form. Good responses simplified the whole expression by first factorising a common factor of n before carrying out further simplification. This generally led to more accurate algebra and made it easier to obtain the required form.

Since the result was not given in the question, recovery from minor errors was permissible, eliminating the most common cause for marks not being awarded in this type of question when it is a 'show that' for a given result.

The most common reason for the last accuracy mark not being awarded was not the summation formulae themselves, but subsequent algebraic manipulation. A few learners successfully reached $n(2n^2 - n)$ or an equivalent intermediate form but did not follow through and final factorisation to $n^2(2n^2 - 1)$ was not seen.

Part (b) was more challenging than part (a). Most learners observed that the given sum was related to part (a) and required the difference between two sums to be found. The limit of 5 in the sum was not correctly applied in the difference of sums, with 6 or even 7 seen used. The question said "hence" (follow through from part (a)) and required the method used to be made clear. This meant that some of the marks were not awarded, even when using the answer to part (a) and just wrote down an answer in this part. One common error was to use incorrect limits such as 100 or 99 for the upper limit and/or either 10 or 12 for the lower limit in the difference of sums.

Question 4

Responses that partitioned the integrand into two fractions were generally successful. A common source of error was the constant associated with the arctangent term, while integration of the logarithmic term was usually handled effectively.

A few learners attempted to integrate by parts in this question, making it a step up in challenge from the preceding questions. This is a theoretically legitimate approach, but it hinges on the ability to integrate the arctan function, which is a result not given in the provided results in the examination. To score any marks via this approach a learner would have to make progress for this integral to reach an analogous form for their integral.

Learners dealt with the mean value requirement well, generally incorporating the 1/5 at the start of the question, though a small number of learners did not use it after integrating, so the final two marks were not awarded.

One notable common error seen repeatedly was expanding at the final step, getting $\frac{1}{5} \left(\frac{1}{2} \ln 2 + \frac{\pi}{4} \right) = \frac{1}{10} \ln 2 + \frac{\pi}{4}$ for which the final A mark was not awarded. Very few used the alternative method, and these had limited success.

Question 5

This question provided accessible marks for many learners, although fully rigorous justification was less common in part (b).

Part (a) was well answered. The most common cause of a mark not being awarded was not using the correct signs when writing the Viète relations. Learners demonstrated a secure understanding of the relationships between the roots and coefficients of a polynomial and were generally able to determine the required expressions with little difficulty.

Part (b) proved to be a more demanding question, testing learners' algebraic skills and their ability to work accurately with quadratic and cubic equations without relying on calculator technology. Most successfully formed either a quadratic or cubic equation involving a single variable. It was more challenging to access the second method mark. Learners used two main approaches to this part, both beginning by expressing each of the three roots in terms of α . One approach used the product of the roots from part (a)(iii) to form a cubic equation in α . Most learners who used this approach went on and obtained a correct cubic equation. The requirements of the question were to show all stages of working even when a calculator is used to solve the equation. This was a reason why some marks were not available. Learners who proceeded to solve

the cubic recognised that $(2\alpha-5)$ was a factor of the cubic, divided to obtain a quadratic equation and then used the discriminant to show that the remaining quadratic had no real roots.

The other main approach used the result from part (a)(ii) together with the expressions for the three roots to obtain a quadratic equation in α . This approach proved to be more successful. Most found both possible values of α , with the principal source of error being to observe which solution should be rejected. Some learners gave both values as valid but may omitted to reject one of them without reference to the conditions given in the question or by checking against the original cubic equation. More successful responses justified the rejected solution by considering the conditions of the question, checking the sign of the product of roots, or substituting into the original equation.

Question 6

This question was answered successfully by many learners, who showed secure understanding of non-homogeneous second order differential equations. Responses generally demonstrated confidence in finding both complementary functions and particular integrals.

In part (a), most learners formed the correct auxiliary equation and obtained the correct complementary function before selecting a suitable particular integral, either as a constant or including a constant term. When an incorrect complementary function was used, it most commonly used 2 instead of 12 in the angle. Some errors occurred when finding the constant in the particular integral, but most obtained the correct value of 30 and hence the correct general solution. The most common incorrect response came from treating the constant in the particular integral as a variable during differentiation, leading to an incorrect value for the particular integral. Less common errors included using incorrect variables (y and x instead of x and t) or omitting $x =$ from the final general solution.

Part (b) required learners to apply the general solution within the context of the model and was generally completed successfully. Most learners correctly interpreted 'initially' as $t = 0$ and recognised that the given information provided initial values for both x and dx/dt . This enabled many responses to determine the required constants and obtain the particular solution. Where errors occurred, these were more often linked to interpreting the model than to solving the differential equation. Common errors included applying the initial condition to the complementary function only, leading to $A = 28$ rather than $A +$

$B = 28$, and treating the given initial rate as though it required conversion from seconds to minutes, giving $\frac{dx}{dt} = \frac{47}{60}$. Other errors included omitting the constant + 30 from the final solution after correctly determining the constants in the general solution, making arithmetic slips when differentiating, making errors when solving for the remaining constant, or not recognising that constants differentiate to zero.

Question 7

This question brought together learners' understanding of complex numbers in exponential form, finite and infinite geometric series, and Euler's formula. Many responses recognised the underlying geometric progression structure, but success depended on accurate manipulation of exponential expressions and clear algebraic communication. Some responses did not follow a clear strategy, particularly in part (i), and therefore did not progress efficiently towards the required form.

The first step in part (i) was to realise that the sum was a geometric progression and to identify its first term, common ratio and the number of terms. Most learners recognised that the given series could be written as a geometric progression with first term $e^{i\theta}$ and common ratio $e^{2i\theta}$ with $n = 10$. The number of terms had the most errors in responses, with often $n=9$ and occasionally 19 seen rather than the correct $n=10$.

The formula for the sum to n terms of a geometric was usually used correctly even with incorrect terms and thus at least one mark was awarded.

Following a correct start, simplifying the resulting expression $S = \frac{e^{i\theta}(1-e^{20i\theta})}{1-e^{2i\theta}}$ presented a greater challenge. Some responses stopped at this stage, while others attempted rewrite the exponential terms using sine and cosine forms, i.e., converting $e^{ni\theta}$ into $\sin(n\theta) + i \cos(n\theta)$, but did not develop a successful strategy. Common errors included using incorrect trigonometric forms, making algebraic slips during conversion, or reaching the printed answer from incorrect working rather than through a valid sequence of steps

Responses that recognised the need to use the exponential form of the sine function were able to make further progress. These responses typically cancelled and factorised terms to form pairs that could be written in terms of $\sin 10\theta$ and $\sin \theta$. As this was a proof, accuracy in the complex form was important. For example, responses that used $2\sin 10\theta$ rather than $2i\sin 10\theta$ did

not gain the final accuracy mark, although benefit of doubt was applied where no technically incorrect step was shown.

The most common errors occurred during the manipulation of exponential terms. Many responses did not successfully rewrite the numerator and denominator in forms involving $e^{i\theta} - e^{-i\theta}$ and $e^{10i\theta} - e^{-10i\theta}$. Errors in factorisation, sign handling and powers of e were often seen, and these frequently prevented responses from gaining the final accuracy mark. Another common error was to multiply both parts of the fraction by $e^{i\theta} - e^{-i\theta}$ or $e^{10i\theta} - e^{-10i\theta}$.

Stronger responses demonstrated a secure understanding of the relationship between exponential and trigonometric forms and were able to manipulate the expression efficiently into the printed result. These responses generally used a clear structure and were less likely to include algebraic slips.

Part (ii) was generally answered more confidently. Many responses identified the infinite geometric series and applied the sum to infinity formula accurately. Where errors occurred, they were usually in the final algebraic simplification of complex exponential fractions.

The first step in part (ii) was to recognise that $P + iQ$ formed an infinite geometric series. This was often seen, followed by correct application of the sum to infinity formula, allowing many responses to access the first three marks efficiently. Most responses first used Euler's formula, although some continued to work directly with sine and cosine forms. A common difficulty was recognising the geometric progression but not explicitly identifying the first term and common ratio.

Once the summation had been set up, the algebraic manipulation required for the simplification was less demanding than in part (i), so correct simplification often followed. Where the final accuracy mark was not gained, this was usually due to errors in simplifying complex exponential fractions. Some responses showed correct intermediate expressions but then included algebraic slips in the final stages.

Question 8

This question tested learners' ability to apply algebraic and calculus techniques within a modelling context. It was attempted by most learners and provided good discrimination across the ability range. Many responses made a successful start in part (a), though fewer maintained complete accuracy throughout the question. More successful responses recognised the links between parts (a) and (c), which often enabled substantial progress in the later stages.

In part (a), the method was familiar to many learners. As this result was not given, there was more scope for credit to be awarded despite minor errors than in questions where a printed result has to be shown. Many responses expanded $\left(z + \frac{1}{z}\right)^7$ in powers of z and then paired terms to use the given identity to obtain terms in $\cos n\theta$. Many responses gained full marks in this part. Where marks were not awarded, this was often because the pairing of terms was not shown clearly, or because notation was not used accurately. For example, some responses used notation such as $\left(z^n + \frac{1}{z^n}\right)^7$ and continued to use powers of n throughout. In some cases, the left-hand side was omitted, so the primary identity was not made clear. As this was a 'show that' question, learners should provide sufficient detail to justify the result. A small number of responses converted all terms into exponential form rather than applying the given result; this approach could still gain most of the marks when carried out accurately.

In part (b) most responses established that k was equal to 8. The most common errors arose from confusion between diameter and radius, leading to answers such as $k = 4$ or $k = 16$.

Performance in part (c) was mixed. After writing down the integral required for the volume of revolution forming the dome and relating it to the answer to part (a), many responses proceeded to integrate correctly. The correct formula was usually applied, although some responses used 2π or omitted π altogether. Many learners successfully used their result from part (a) to write $\cos^7 y$ in a form suitable for integration, and responses with a correct expression in part (a) were often able to score highly in this part, even where minor algebraic slips occurred later.

Some responses attempted to use a reverse chain rule method, for example via $\cos^2 \theta = 1 - \sin^2 \theta$. This approach could be worked through successfully, but it tended to be more prone to error. Many learners did not change the variable from θ to y in their integral; this was not penalised, so these responses could still gain the marks available for the integral.

Incorrect attempts at the integration, such as going to expressions $\cos^8 y$, were also seen frequently in less successful responses. Learners should follow through their result in part (a) and use it in part (b).

Another common error in this part involved the limits of integration, with some using limits such as 0 and 2 or 0 and 8, confusing the x and y axes.

In finding the total volume of the paperweight the volume of the cylinder was sometimes omitted. Also seen was an incorrect formula used for the cylinder

volume with 16 used as the diameter or even using the area of the rectangular cross-section for the volume.

This part of the question was testing the ability to round an answer appropriately, so a precise answer was required. The answer had to be rounded to the nearest cm^3 . This was usually well done with only a few being not awarded a mark for accuracy, and this usually due to leaving the answer in an exact form rather than translating into decimal form as is appropriate when modelling. Units were not required but if they were included, they had to be correct.

Though part (d) was generally well answered, learners still find commenting on a model challenging. It was expected that there would be comments about the appropriateness of the model's shape relating to the real paperweight. For example, the equation of the curve might not fit the dome, the measurements might not be accurate, and the paperweight might not be smooth.

Unsuccessful attempts did not relate the model to the paperweight and just commented on features of the paperweight included the paperweight is not solid and the thickness of the paperweight has not been considered, while some responses suggested that the paperweight was hollow. Handwriting was also difficult to make out in some responses, though keywords could mostly be identified. However, reference to paper rather than to the paperweight, for instance, would lose the mark.

Question 9

This question, whilst the final question in the paper, proved to be accessible for most learners, and scored well. Where errors arose, they tended to be in the latter parts of the question.

Part (a) was generally answered successfully, with most responses showing the required formation of the equation accurately. Where marks were not gained, this was usually due to algebraic slips or insufficient working.

Part (b) began to discriminate between responses. Many learners solved the equation correctly and set up the correct form of the general solution. However, some responses then made errors when forming the two equations in A and B needed to determine the constants, most commonly by not using the given conditions to form the second equation. These errors affected the marks available in this part and in later parts of the question.

Most responses successfully formed two correct equations and solved for the constants. The common approach was to differentiate x first to obtain an expression for y in terms of A and B , then use the initial conditions to obtain the equations needed for both (b) and (c) together. Responses that used the

conditions to get the initial value for $\frac{dx}{dt}$ and form the second equation were usually successfully and obtained the correct equation for y in part (c). Some numerical errors were seen when finding the coefficients in the differential equations including setting $\frac{dx}{dt}$ to 0 instead of -14.

Part (d) was generally well answered. In part (i), the follow-through mark meant that many responses could gain the mark even after errors in parts (b) and (c). The most common error was using $t = 15$ instead of $t = \frac{1}{4}$ in their equation for x .

Responses with correct equations generally solved part (d)(ii) successfully. Where earlier errors had led to equations that were more difficult to solve, some responses were not able to access these marks. The logarithmic work was usually carried out well, although some errors in manipulating logarithmic and exponential expressions were seen. Many final answers were given in decimal form after $t = \ln\left(\frac{15}{4}\right)$, but the mark was awarded when the exact value had been seen. Responses giving $t = -\ln\left(\frac{4}{15}\right)$ also gained the mark. The most common error in this part was to use $x = 4y$ rather than the required $y = 4x$.

Part (e) was an effective discriminator at the end of the paper. Many responses gave the correct conclusion that the model was not suitable but did not always provide a sufficiently rigorous mathematical justification. For example, statements such as “1.32 is not close to 2” needed to be supported by a comparison of relative size or a percentage difference.

Some responses identified that the expressions became negative, which was sufficient to justify that the model was no longer viable. This was not commonly seen, but responses that made this observation did not need to provide further reasoning.

To justify the conclusion fully, learners needed to provide a mathematical basis for their decision. This was most seen through a percentage error calculation, demonstrating that the relative error of the model was greater than 30%. Learners should be encouraged to justify whether a model is appropriate by using clear mathematical evidence to support their conclusion. A small number of responses identified that the expressions became negative, which was sufficient to show that the model was no longer suitable. In these cases, no further reasoning was needed, as this demonstrated that the model was no longer viable. This approach was not commonly seen, as many responses did not check the relevant values. Where this was identified, it was often used successfully as the justification. Although this observation may have led some learners to review their earlier work, there was little evidence that it affected progress through the question, as it appeared only in the final part of the paper.